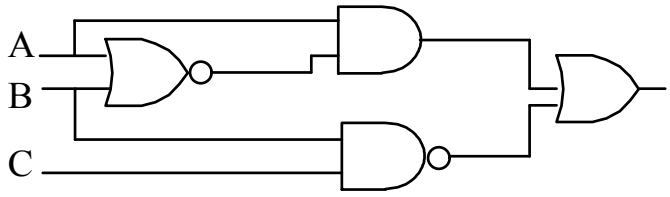
Compiled Problems – Intermediate Division

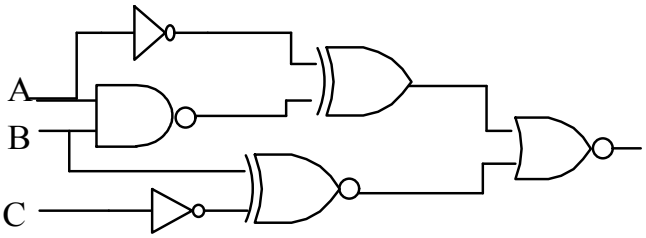
1. **00-01 C3 Digital Electronics**

   Find all ordered triples that make the circuit **FALSE.**



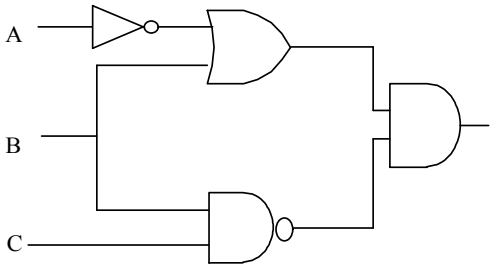
2. **00-01 C3 Digital Electronics**

   Write a Boolean Algebra expression that is an exact translation of the following circuit. Do not simplify the expression and only use parentheses when required.



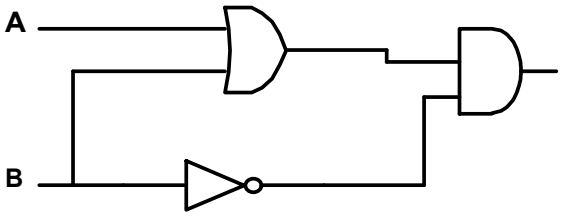
3. **00-01 C4 Digital Electronics**

   How many triples make the following circuit TRUE?



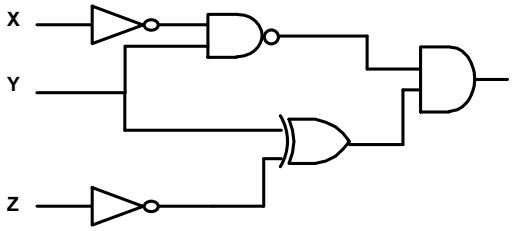
4. **01-02 C3 Digital Electronics**

   Find all ordered pairs (A,B) that make the following circuit TRUE.

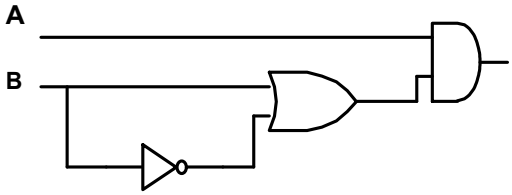


5. **01-02 C3 Digital Electronics**

Translate the circuit to a Boolean expression and simplify.



6. **01-02 C4 Digital Electronics**

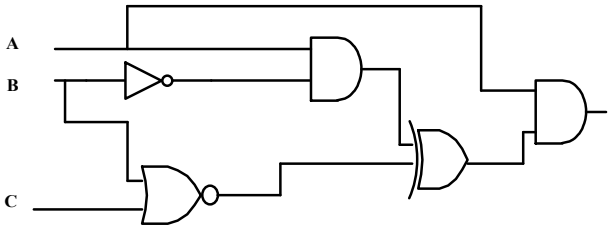


List all of the following statements that are true about the circuit above.

a. It is TRUE for all possible inputs.
b. It is FALSE for all possible inputs
c. It is TRUE only when the two inputs are different.
d. It is sometimes TRUE when the two inputs are different.
e. It is FALSE whenever B is TRUE.

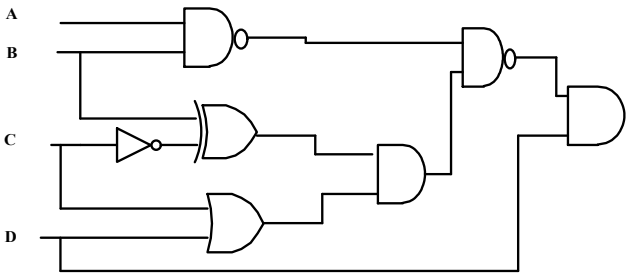
7. **02-03 C3 Digital Electronics**

List all ordered triples that make the following circuit TRUE:



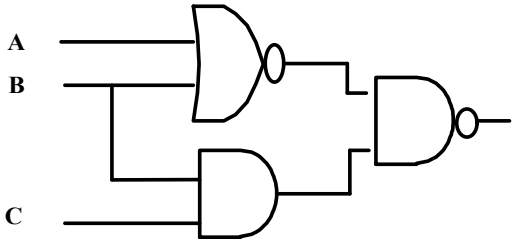
8. **02-03 C3 Digital Electronics**

Write a Boolean expression that translates the following circuit.
Do not simplify and only use parentheses when required.



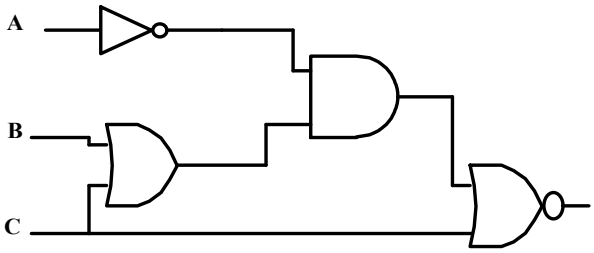
9. **02-03 C4 Digital Electronics**

How many ordered triples make the following circuit FALSE?



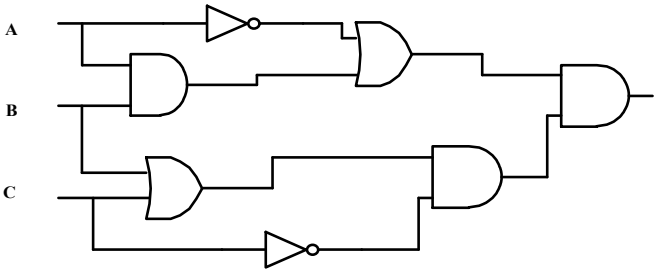
10. **03-04 C3 Digital Electronics**

Find all ordered triples that make the following circuit TRUE:



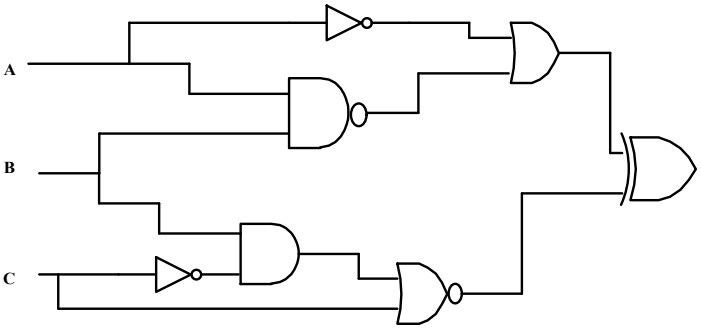
11. **03-04 C3 Digital Electronics**

Convert the following diagram to a Boolean expression and simplify



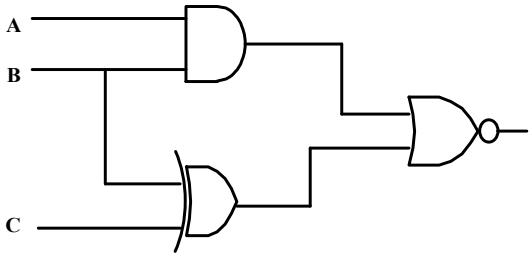
12. **03-04 C4 Digital Electronics**

List all ordered triples that make the following circuit TRUE.



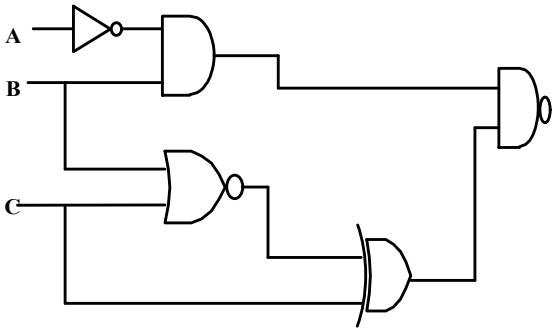
13. **04-05 C3 Digital Electronics**

List the ordered triples that make the following circuit TRUE.



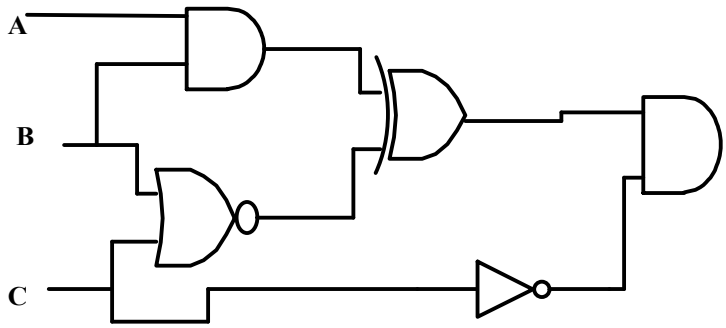
14. **04-05 C3 Digital Electronics**

Translate the circuit. Do not simplify. Use the minimum number of parentheses.



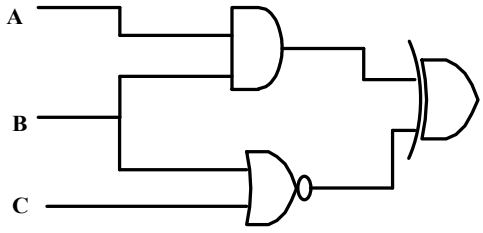
15. **04-05 C4 Digital Electronics**

Which ordered triples make the following circuit TRUE?



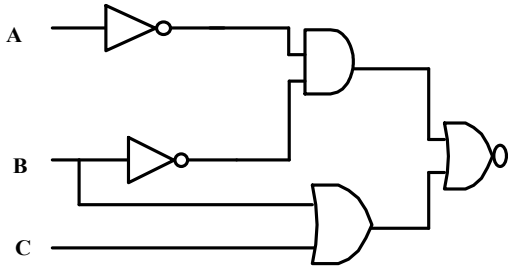
16. **05-06 C3 Digital Electronics**

How many ordered triples make the following circuit TRUE?



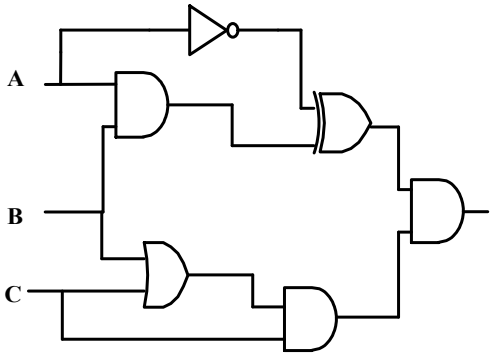
17. **05-06 C3 Digital Electronics**

Translate the following circuit to a simplified Boolean expression.



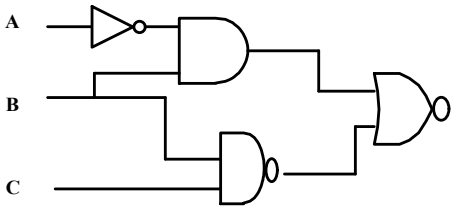
18. **05-06 C4 Digital Electronics**

List all the ordered triples that make the circuit TRUE.



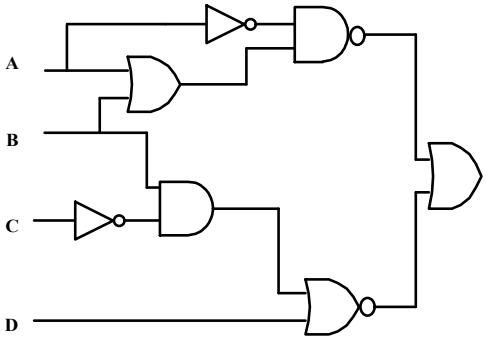
19. **06-07 C3 Digital Electronics**

Convert this circuit to a Boolean expression and simplify.



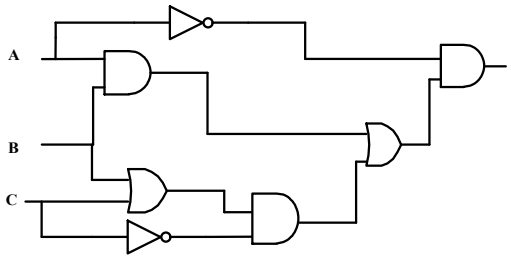
20. **06-07 C3 Digital Electronics**

How many ordered 4-tuples make this circuit FALSE?



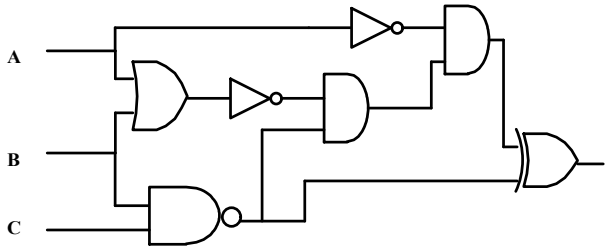
21. **06-07 C4 Digital Electronics**

Convert the digital diagram to a Boolean expression and simplify.



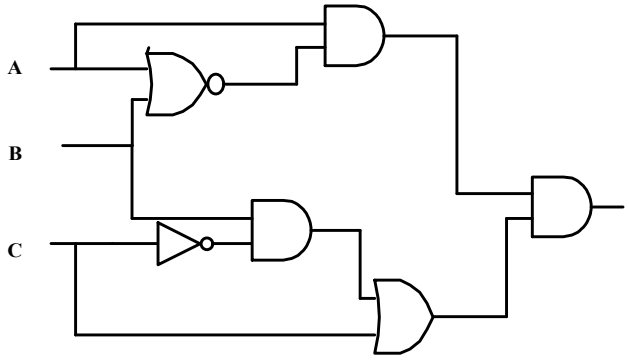
22. **07-08 C3 Digital Electronics**

Which ordered triples make this circuit FALSE?



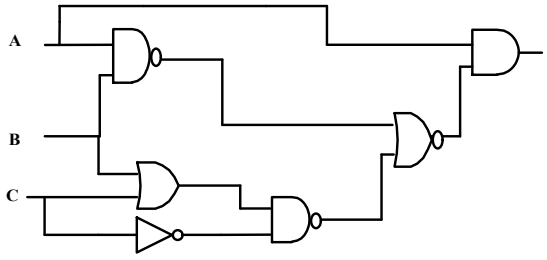
23. **07-08 C3 Digital Electronics**

Simplify the Boolean expression that this circuit represents.



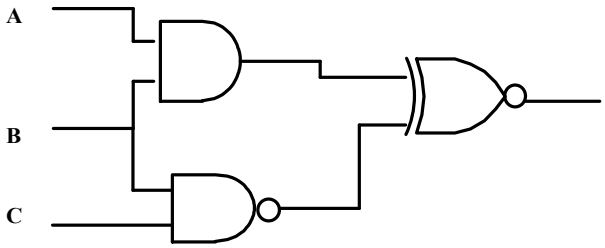
24. **07-08 C4 Digital Electronics**

List all the triples that make the digital circuit TRUE.



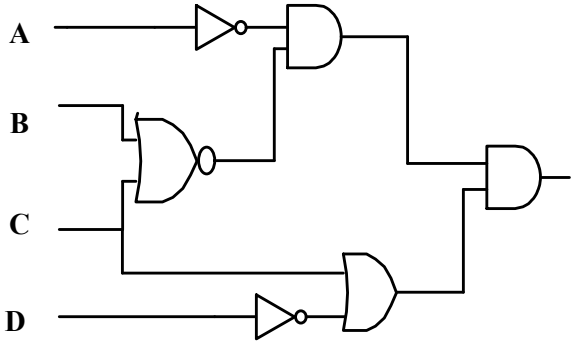
25. **08-09 C3 Digital Electronics**

How many ordered triples make this circuit TRUE?



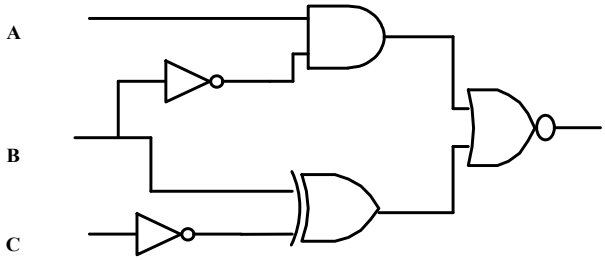
26. **08-09 C3 Digital Electronics**

Simplify the boolean expression that is represented by this circuit.



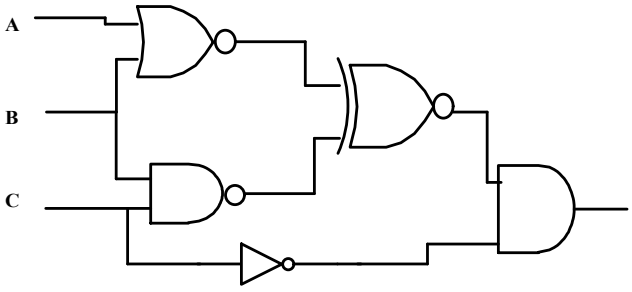
27. **08-09 C4 Digital Electronics**

List all the triples that make the digital circuit TRUE.



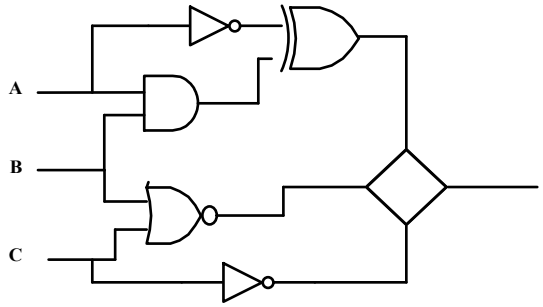
28. **09-10 C3 Digital Electronics**

Which ordered triples make this circuit TRUE?



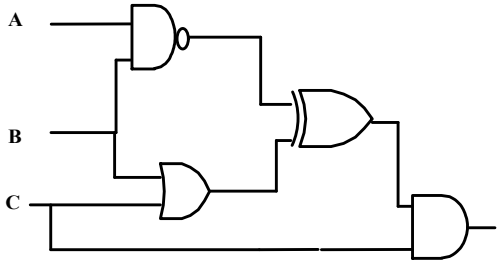
29. **09-10 C3 Digital Electronics**

Define a new gate, diamond, as TRUE if only 1 of its three inputted values is TRUE. Which ordered triple(s) make the circuit TRUE?



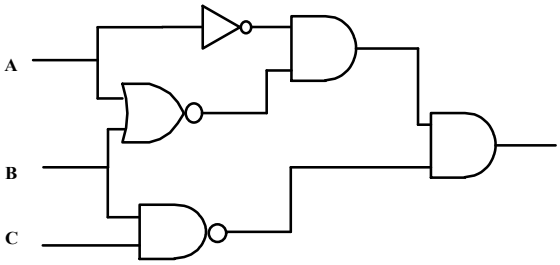
30. **09-10 C4 Digital Electronics**

List all the triples that make the digital circuit TRUE.



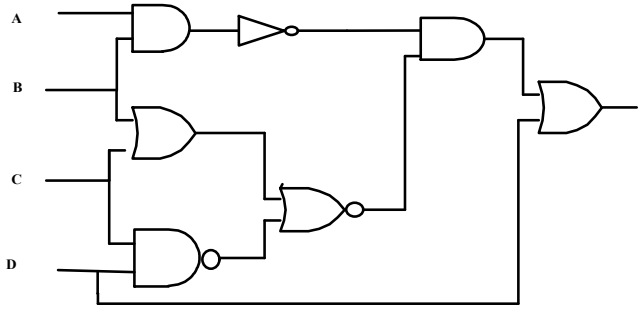
31. **10-11 C3 Digital Electronics**

How many ordered triples make this circuit TRUE?



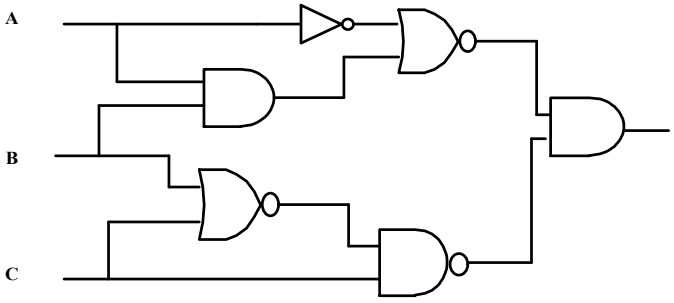
32. **10-11 C3 Digital Electronics**

Simplify the Boolean expression that is represented by this circuit.



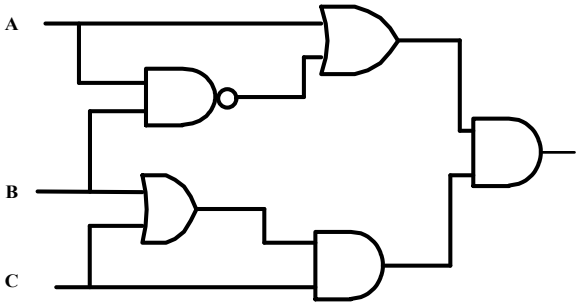
33. **10-11 C4 Digital Electronics**

How many triples make the digital circuit TRUE?



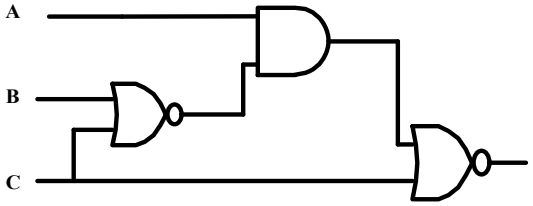
34. **11-12 C4 Digital Electronics**

Simplify the Boolean expression that this circuit represents.



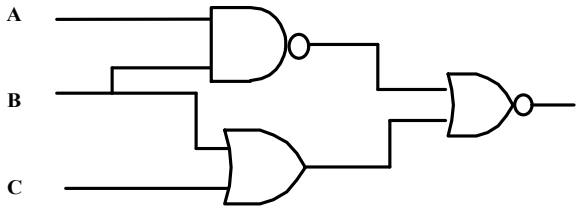
35. **11-12 C4 Digital Electronics**

Which ordered triples make the digital circuit TRUE?



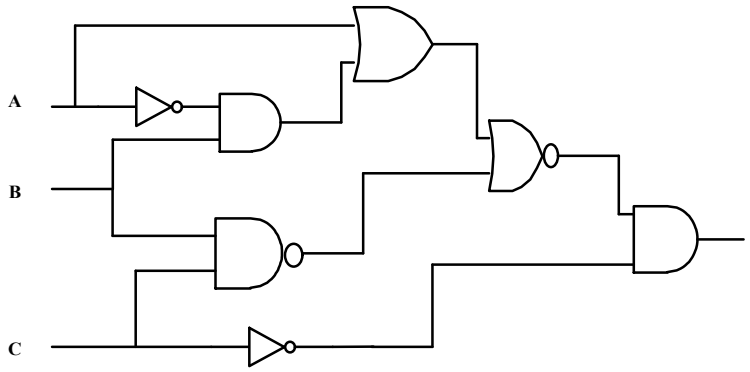
36. **12-13 C4 Digital Electronics**

What ordered triple(s) make the following circuit TRUE?



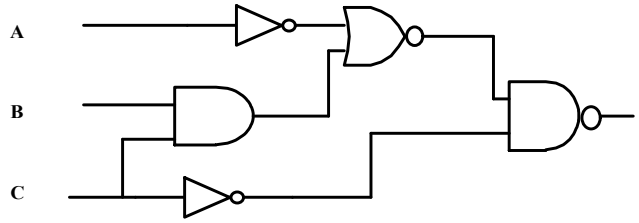
37. **12-13 C4 Digital Electronics**

How many ordered triple(s) make the circuit TRUE?



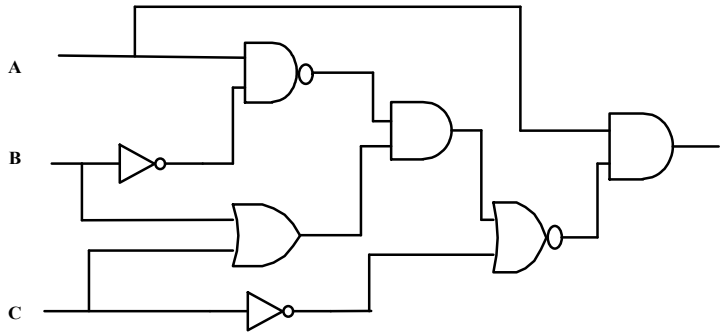
38. **13-14 C4 Digital Electronics**

What ordered triple(s) make the following circuit FALSE?



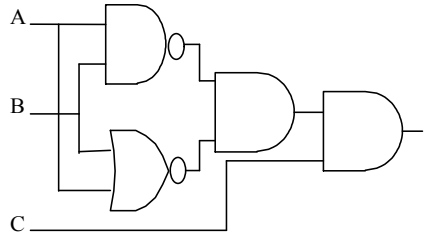
39. **13-14 C4 Digital Electronics**

Simplify the Boolean expression represented by the circuit below.



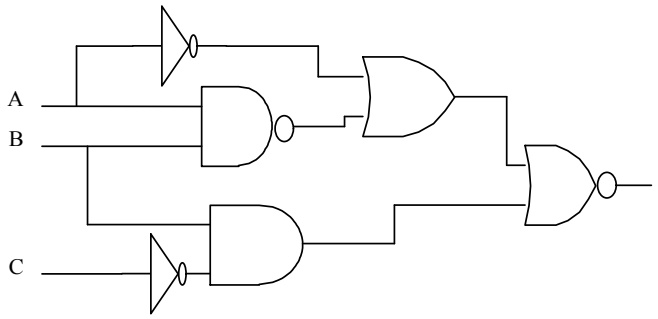
40. **14-15 C4 Digital Electronics**

Translate the following circuit to a Boolean expression and simplify the expression.



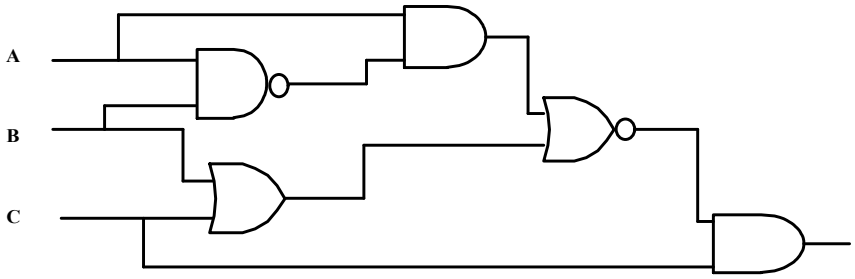
41. **14-15 C4 Digital Electronics**

List all the ordered triples that make the circuit TRUE



42. **15-16 C4 Digital Electronics**

How many ordered triples make the circuit FALSE?



43. **15-16 C4 Digital Electronics**

Simplify the Boolean expression represented by the circuit below.

